

1. INTRODUCTION

Brain tumors represent one of the most critical and life-threatening medical challenges faced today. These abnormal growths result from uncontrolled cell division within the brain or nearby structures, leading to severe functional impairments. Even small tumors can have dangerous consequences because of the limited space within the skull and the high sensitivity of neural tissues. Tumors may compress vital brain regions responsible for movement, memory, speech, emotion, and vision, creating a wide range of neurological symptoms. Given the complexity of the human brain, early identification and monitoring of tumors play an essential role in ensuring timely intervention and improving patient survival rates. Medical imaging has become indispensable in this context, with Magnetic Resonance Imaging (MRI) serving as the primary diagnostic tool due to its superior soft tissue contrast and ability to reveal even minute abnormalities without exposing patients to harmful radiation.

Despite the strength of MRI technology, the process of interpreting MRI scans remains challenging. Radiologists must manually analyze several slices of 2D or 3D brain images, carefully examining textures, intensities, and structural differences to detect abnormalities. This task becomes even more difficult when tumor boundaries are blurred or when tumor intensity closely resembles that of surrounding tissues. MRI images also differ significantly based on the type of scanner, field strength, imaging sequence, and patient movement during the procedure. Variations in brightness and contrast can obscure small tumors or make larger tumors appear less distinct. These challenges often require a radiologist to rely heavily on experience, pattern recognition skills, and prolonged visual examination. In busy clinical environments, radiologists must handle many cases each day, which increases the likelihood of fatigue-induced errors. Even highly trained specialists may occasionally miss subtle lesions or misinterpret ambiguous regions.

The last decade has witnessed significant advancements in Artificial Intelligence (AI), particularly in the field of medical imaging. AI systems have shown an impressive ability to identify hidden patterns in complex images that may not be easily noticeable to the human eye. Deep learning models, especially Convolutional Neural Networks (CNNs), are capable of processing thousands of MRI slices and extracting meaningful features automatically. CNNs have been used successfully in numerous medical applications, including disease detection, tumor classification, organ segmentation, and biometric analysis. Their hierarchical structure

enables them to learn simple features like edges and intensities at shallow layers and progressively learn more complex patterns like textures, shapes, and tumor-specific characteristics at deeper

layers. This ability to learn directly from raw images without human-designed features gives CNNs a major advantage over classical machine learning techniques.

However, most existing deep learning models for brain tumor diagnosis focus solely on classification—determining whether a tumor exists and identifying its type. While classification is important, it is only one aspect of clinical diagnosis. Doctors also need to know where the tumor is located and how large it is to plan surgical interventions or radiotherapy. A model that only classifies the tumor and ignores segmentation provides limited clinical usefulness. Tumor boundaries must be clearly identified for precise surgical removal and monitoring tumor progression over time. Therefore, a fully functional diagnostic system requires both accurate segmentation and classification.

This project proposes a hybrid model that integrates fuzzy thresholding with CNN-based classification to address these limitations. Fuzzy thresholding is an intelligent image segmentation technique inspired by fuzzy logic. Unlike traditional thresholding that separates pixels into binary classes using a fixed threshold, fuzzy thresholding assigns a degree of membership to each pixel. This makes it highly suitable for MRI images, where pixel intensities vary gradually, and boundaries are rarely sharp. Tumor regions often share visual similarities with normal brain tissue, making crisp segmentation unreliable. Fuzzy thresholding evaluates uncertainty and produces a more refined, accurate tumor mask, capturing the subtle differences in intensity and texture. By applying fuzzy thresholding, this project ensures that the CNN classifier receives cleaner and more precise input data.

Once segmentation identifies the tumor region, the CNN model classifies the MRI image into four categories: Glioma, Meningioma, Pituitary Tumor, or No Tumor. Each of these tumor types has unique visual characteristics and clinical implications. Gliomas, for example, arise from glial cells and often infiltrate deep brain tissues, making them more difficult to remove surgically. Meningiomas grow along the outer surface of the brain and are usually non-cancerous, but their size and location can still cause significant pressure on critical brain structures. Pituitary tumors develop near the pituitary gland and may affect hormone production, having widespread effects on the body. Identifying the correct tumor type helps doctors select an appropriate treatment strategy, making accurate classification a crucial part of the diagnostic process.

An additional novel aspect of this project is the estimation of tumor size. Calculating the tumor's area relative to the entire brain region is extremely useful for evaluating tumor aggressiveness, monitoring growth, and assessing treatment response. Many existing AI systems do not perform

size estimation, leaving another manual task for radiologists. This project uses the segmented tumor mask from fuzzy thresholding to compute the number of tumor pixels and compare them with the total number of pixels representing the brain region. The resulting percentage gives a quantitative measure of tumor size, which can be important when tracking tumor progression over time.

One of the major reasons for building an automated detection system is the increasing workload on radiologists. Hospitals often handle hundreds of MRI scans daily, and radiologists must provide quick and accurate interpretations for each one. Automated AI tools can help reduce this burden by pre-processing and pre-evaluating MRI scans, highlighting suspicious regions before the radiologist reviews the images. Such systems do not replace doctors but serve as supportive tools to improve accuracy, reduce errors, and speed up the diagnostic process. Furthermore, in rural and underdeveloped regions where experienced radiologists may not be available, AI-based tools can support telemedicine services by providing preliminary analysis so that remote doctors can make informed decisions.

This project is motivated by the need to improve diagnostic accuracy, reduce manual workload, and create a scalable system that can eventually be integrated into clinical practice. The hybrid approach combining fuzzy thresholding and CNN is designed to address real-world challenges in MRI interpretation, such as low contrast, intensity variations, irregular tumor shapes, and limited availability of annotated datasets. The ultimate goal is to create a reliable, automated system that can detect, segment, classify, and estimate tumor size with high accuracy and minimal human intervention.

The project involves several key stages, beginning with collecting publicly available MRI datasets that include different tumor types as well as normal brain scans. Preprocessing techniques such as resizing, normalization, and data augmentation are applied to standardize the images and enhance model robustness. Fuzzy thresholding is then used to isolate the tumor region from the background. Tumor size is calculated by analyzing the number of pixels identified as part of the tumor. A CNN model is trained to classify MRI images into their respective categories, and its performance is evaluated using accuracy, precision, recall, and F1-score. Finally, the results are compared with existing approaches to highlight improvements achieved by the proposed method.

In addition to the increasing demand for automated diagnostic tools, another challenge faced in brain tumor detection is the inconsistency that arises due to differences in datasets. MRI images collected from different hospitals, scanners, and imaging conditions often vary in quality and

intensity distribution. A model trained on one dataset may not perform equally well on another unless it is designed to handle such variations. This makes it important to adopt techniques like fuzzy thresholding, which can adapt to dynamic changes in image intensity and perform reliable segmentation even in challenging scenarios. By incorporating fuzzy logic principles, the system becomes more tolerant of noise, uneven illumination, and complex texture patterns commonly found in real-world MRI scans.

A major concern in today's medical field is the increasing number of patients requiring intensive diagnosis while the number of trained radiologists remains relatively stagnant. This gap creates delays in diagnosis, extended waiting times, and, in some cases, late detection of critical diseases. Artificial intelligence supports the medical workforce by providing immediate pre-analysis of MRI images, enabling doctors to prioritize urgent cases. Such tools do not replace radiologists but enhance their ability to make quicker and more accurate decisions. Integrating automated tumor identification and classification can significantly reduce the chances of human oversight and improve patient outcomes, especially in emergency situations where time is crucial.

Another driving factor for developing this hybrid model is the complexity of tumor morphology. Brain tumors vary greatly in shape, size, and internal structure. Some tumors are compact and well-defined, while others infiltrate the surrounding tissues, making segmentation extremely difficult. Traditional image processing techniques may struggle with such irregularities. The combination of fuzzy thresholding and deep learning equips the system with both interpretability and precision. Fuzzy segmentation provides an approximate but meaningful boundary, while the CNN learns detailed structural differences that distinguish tumor types. Together, these methods create a more comprehensive analysis pipeline that mimics the multi-step reasoning used by human experts.

Over time, deep learning has evolved from simple neural network structures to advanced architectures capable of handling millions of images. CNN models have revolutionized image classification by using convolutional filters that capture essential features from different levels of abstraction. These filters help the model detect subtle differences in texture and intensity that are not easily identifiable through conventional techniques. In medical imaging, where every small detail can indicate a significant change, such learning capability becomes extremely valuable. The integration of CNNs in this project ensures that classification results are not only accurate but also robust against variations in tumor appearance.

The hybrid approach used in this project also supports scalability. Once the system is trained, it

can process a large number of MRI images in a very short time, something that would be nearly impossible for humans to achieve manually. Hospitals and diagnostic centers dealing with heavy patient loads can benefit from such an automated solution, especially during peak times or in regions with limited medical staff. The proposed system can be further enhanced in the future to provide additional insights such as tumor growth prediction, survival analysis, and personalized treatment recommendations. These potential extensions underscore the importance of developing a strong and reliable foundation, which begins with effective segmentation and classification.

An essential part of this project is the focus on accuracy and real-world applicability. Unlike traditional academic studies that concentrate mainly on model performance metrics, this research aims to create a practical diagnostic tool suitable for deployment in clinical environments. By estimating tumor size, the system goes beyond simple classification and offers measurements that can support oncologists in planning surgeries, assessing treatment response, and predicting disease progression. Tumor size is a key factor in medical decision-making; therefore, its automated computation reduces workload and reduces the risk of human error.

Furthermore, the proposed system aligns with the growing initiative toward digital healthcare and intelligent medical technologies. Countries around the world are investing heavily in AI-driven medical solutions to improve disease detection and reduce healthcare costs. Automated analysis tools are particularly useful in telemedicine, where remote diagnosis is often needed for patients in rural or underserved areas. A system capable of analyzing MRI scans accurately can bridge the gap between patients and healthcare providers, ensuring timely diagnosis without requiring specialized equipment or experts to be physically present.

The development of this system also required addressing ethical and reliability concerns. In medical applications, accuracy alone is not sufficient; the model must be stable, interpretable, and safe to use. While deep learning models are often criticized for being “black boxes,” the use of fuzzy thresholding provides an element of interpretability by showing how tumor boundaries are generated. This makes the diagnostic process more transparent and easier for doctors to trust. Ensuring that the system provides consistent performance across different datasets and imaging conditions further strengthens its reliability. These factors contribute to building a trustworthy AI system suitable for clinical integration.

The significance of this project lies not only in its technical contribution but also in its potential to improve human lives. Brain tumor diagnosis is a sensitive and crucial area of healthcare where timely detection can save lives. A system that reduces diagnosis time, improves accuracy, and

assists doctors has far-reaching impacts. Patients can receive earlier treatment, radiologists can focus on more complex tasks, and hospitals can manage resources more efficiently. As AI becomes increasingly integrated into healthcare, projects like this pave the way for safer, smarter, and more accessible medical diagnostics.

Overall, this introduction highlights the significance of early tumor detection, the challenges associated with manual MRI interpretation, and the advantages of combining fuzzy logic with deep learning techniques. By providing an automated system capable of segmentation, classification, and size estimation, this project offers a meaningful step toward enhancing brain tumor diagnosis and supporting clinical decision-making. The hybrid approach not only improves accuracy but also contributes to the advancement of intelligent healthcare systems, paving the way for future innovations in medical imaging and AI-assisted diagnostics.

2.LITERATURE REVIEW

Introduction of Literature Review

The field of brain tumor detection has undergone significant transformation over the past two decades, primarily due to advancements in medical imaging and computational intelligence. Early research efforts relied heavily on manual inspection of MRI scans by experienced radiologists. While MRI technology provided clear visualization of brain tissues, interpreting subtle abnormalities remained a subjective process influenced by the radiologist's experience, fatigue, and variations in image quality. As medical imaging datasets grew in size, it became increasingly evident that manual diagnosis alone was insufficient for handling complex cases efficiently, motivating researchers to explore automated computational techniques capable of assisting clinicians. These initial attempts laid the foundation for several image-processing and machine-learning approaches that have continued to evolve with time.

After this, machine-learning techniques like SVM, KNN, ANN, and Decision Trees were explored. These models used manually extracted features such as texture and shape. They gave better results than classical techniques, but the major drawback was that handcrafted features often missed subtle tumor patterns. This led to inconsistent performance, especially when dealing with unfamiliar datasets. The biggest improvement came with the introduction of deep learning, especially Convolutional Neural Networks (CNNs). CNNs can automatically learn important features from images without manual intervention, and several studies reported very high classification accuracy, often above 95%. CNNs became very effective for identifying tumor types like glioma, meningioma, and pituitary tumors. However, most deep-learning models focused only on classification and did not perform segmentation or tumor size measurement, which are important for real clinical use.

When we look at the existing literature on brain tumor detection, we can clearly see how the field has evolved with advancements in imaging and artificial intelligence. In the early years, most diagnosis depended entirely on radiologists who manually inspected MRI scans. Although radiologists are highly trained, manual interpretation takes time and may differ from person to person. Many studies pointed out that MRI scans often contain noise, varying brightness levels, and unclear tumor boundaries, which make manual segmentation difficult. Because of these issues, researchers started developing automated methods that could support radiologists and reduce human error.

One of the first automated approaches used in the literature was threshold-based segmentation. This method separates the tumor from the background based on pixel intensity values. While simple, researchers soon realized that thresholding works only when the tumor has a very different intensity compared to normal tissue. But in real MRI images, tumor intensities often overlap with brain tissues. Several papers reported that this method fails when tumor boundaries are irregular or when the image contains noise. These early studies highlighted the need for more adaptable techniques.

After traditional methods, machine-learning algorithms started gaining attention. Models like SVM, KNN, ANN, and Decision Trees were widely tested for tumor classification. These models required manually designed features such as texture, shape, and histogram values. Many research works achieved moderate accuracy using these handcrafted features, but they also pointed out a major drawback: manually selected features may not capture all the hidden details inside MRI images. If the dataset changes or contains more complex tumor shapes, the model performance drops. So even though machine learning was a step forward, it still had limitations in terms of generalization and robustness.

A major shift occurred when fuzzy logic-based segmentation was introduced in the literature. Fuzzy methods were designed to handle uncertainty, which is very common in medical images. Unlike classical thresholding, fuzzy thresholding does not force pixels into strict categories. Instead, it gives each pixel a membership value, deciding how much it belongs to tumor or non-tumor regions. Studies showed that fuzzy logic performs well when tumor boundaries are unclear or when images have brightness variations. This adaptability made fuzzy thresholding a strong candidate for tumor segmentation. Several researchers demonstrated that fuzzy methods outperform normal thresholding because they can handle overlapping pixel intensities and provide smoother segmentation results.

The biggest advancement in the literature came with the rise of deep learning, especially Convolutional Neural Networks (CNNs). CNNs became extremely popular because they can automatically learn the best features from MRI images without any manual effort. Many research studies reported classification accuracies above 95%, which is much higher than traditional machine-learning algorithms. CNNs have the ability to detect subtle features that human-defined features cannot capture. Deep-learning-based models like VGGNet, ResNet, AlexNet, and Inception have been applied in brain tumor classification and have shown remarkable success. However, most CNN-based research focuses only on classification and does not identify the exact

location of the tumor or its size, limiting its usefulness for clinical decision-making.

Several studies also discussed the key limitations in previous systems. Many existing works did not perform accurate segmentation before classification. Without segmentation, the model may misinterpret normal brain tissue as part of a tumor. Another important drawback found in the literature is the absence of tumor size estimation. Size is an important factor used by doctors to plan surgeries and treatments, yet only a few systems calculate it. A common issue highlighted by researchers is that many studies use small or single-center datasets, meaning the models cannot generalize well when used on MRI scans from different hospitals. Variations in brightness, orientation, and noise levels also cause inconsistent performance in several older systems.

Because of these limitations, researchers started exploring hybrid models that combine fuzzy logic with deep learning. Many recent papers show that fuzzy thresholding helps in extracting the tumor region more accurately, while CNN improves classification performance. This combination makes the system more reliable, especially in complex MRI images where boundaries are not clear. Literature also shows that hybrid methods reduce false detections and improve the clarity of segmented tumor regions. Several researchers suggested that combining segmentation and classification in a single pipeline offers better clinical relevance compared to using only classification models.

Another important observation in the literature is the increasing use of publicly available datasets like BraTS, Figshare, and Kaggle MRI datasets. These datasets help researchers compare their models and evaluate performance across different conditions. However, multiple studies mentioned that dataset imbalance is still a major issue. For example, some tumor types may have fewer samples, which affects the training process. Researchers have proposed data augmentation, preprocessing, and normalization techniques to reduce this imbalance and make deep-learning models more robust.

Recent literature also emphasizes the importance of computational efficiency. While CNNs are highly accurate, they sometimes require powerful hardware to process large MRI datasets. This limits their use in smaller hospitals or rural healthcare centers. Researchers have pointed out the need for lightweight models that can work efficiently without requiring expensive GPUs. Papers also highlight the importance of developing systems that can function in real time and assist doctors during diagnosis.

From the overall review of existing research, it becomes clear that no single technique is sufficient to achieve complete brain tumor detection. Traditional methods perform poorly on complex MRI images, machine-learning models depend too much on manual features, and CNNs alone do not provide segmentation or size estimation. Fuzzy logic offers strong segmentation capabilities but needs to be combined with a powerful classifier for full diagnosis. These gaps in the literature strongly support the development of a hybrid system that includes detection, segmentation, classification, and tumor size estimation—all within a single integrated model.

The literature collectively shows that fuzzy thresholding enhances segmentation accuracy, CNN improves classification performance, and tumor size estimation adds clinical value. Combining these strengths helps create a more complete diagnostic tool, which is what our project aims to achieve. This review ultimately highlights the significance of hybrid AI methods and the growing need for automated, accurate, and reliable systems for brain tumor detection using MRI images.

Over the years, brain tumor detection has evolved from basic manual interpretation to highly sophisticated computational methods, each addressing different challenges associated with MRI analysis. The growing availability of digital medical imaging and publicly accessible datasets has significantly accelerated research in this field. MRI is widely regarded as the most reliable imaging technique for identifying brain abnormalities due to its exceptional ability to capture internal structures with fine detail. However, as tumor appearance varies greatly depending on the type, stage, and growth pattern, automated analysis of MRI images has remained a major research focus. The literature highlights that brain tumors often exhibit irregular shapes, heterogeneous textures, and overlapping intensity patterns with surrounding tissues, making automated detection a complex task. These challenges have motivated researchers to explore advanced segmentation and classification algorithms capable of handling such variability.

A large body of literature focuses on understanding the biological and structural characteristics of different tumor types, including gliomas, meningiomas, and pituitary tumors. Gliomas, for instance, are known for their highly infiltrative nature, often blending into healthy tissues, which makes segmentation extremely difficult. Researchers observed that traditional image-processing methods fail to handle such complexity. Meningiomas, on the other hand, are generally well-

circumscribed but still vary in intensity depending on MRI modality. Pituitary tumors occur in the pituitary gland, located at the base of the brain, where imaging is complicated by surrounding bone

and hormonal

variability. Understanding these variations is essential for developing automated systems, and this has been emphasized repeatedly across the literature. Many studies

argue that any effective detection method must incorporate adaptive thresholding or learning-based techniques capable of processing complex texture and intensity patterns.

Another major area discussed in the literature is image preprocessing, which plays a crucial role in improving detection accuracy. MRI images often suffer from noise, intensity non-uniformity, motion artifacts, and spatial distortions. Several studies have proposed preprocessing techniques such as skull stripping, histogram equalization, Gaussian filtering, and intensity normalization to improve image quality. A recurring theme across research articles is that segmentation and classification performance heavily depends on the quality of preprocessing. Some studies show that even a simple normalization step can dramatically improve CNN accuracy, while others highlight that fuzzy-based preprocessing helps enhance contrast between tumor and healthy tissue. Literature also reports that preprocessing reduces the computational burden on classification networks, allowing models to focus on meaningful image regions rather than noise.

Segmentation is one of the most extensively studied tasks in brain tumor research. Several notable approaches have been proposed, including region-based, clustering-based, hybrid models, and deep-learning architectures. Region-based segmentation techniques like region growing or active contour models are effective when tumor boundaries are distinguishable, but they struggle with complex textures and unclear boundaries. Clustering methods like k-means and fuzzy c-means (FCM) have gained popularity due to their ability to categorize pixels into different clusters based on similarity. However, the literature clearly indicates that fuzzy-based clustering outperforms hard clustering in medical imaging, as it accommodates uncertainty and overlapping intensities. Hybrid methods combining fuzzy logic with watershed algorithms or morphological processing have also reported promising results. The literature consistently emphasizes that fuzzy approaches show superior robustness when MRI images have low contrast or noise, making them more suitable for tumor segmentation in real-world conditions.

Classification has also been widely researched in the literature, with CNNs being the most

successful models for this task. Early works used shallow neural networks, but they quickly became outdated as deeper architectures offered superior performance. CNNs such as AlexNet, VGG16,

ResNet50, and DenseNet121 have been applied for tumor classification, and many studies reported accuracies above 95%. Researchers highlighted that deeper models capture more complex features, allowing them to distinguish subtle intensity variations across tumor types. Transfer learning models trained on large datasets like ImageNet have also demonstrated impressive results on medical datasets, despite differences between natural and medical images. The literature also explores ensemble methods that combine multiple CNN models to further enhance classification reliability.

However, despite their success, most CNNs do not include segmentation or localization, limiting their clinical usefulness.

3.OBJECTIVES OF THE PROJECT

Another important objective of the project is to create a diagnostic system that ensures high reliability even under challenging imaging conditions. MRI scans differ significantly depending on the machine type, magnetic field strength, slice thickness, patient movement, and scanning protocol. These variations often cause distortions and inconsistencies that can mislead automated systems. Therefore, the project emphasizes developing a robust model that can adapt to diverse MRI characteristics without producing false detections or misclassifications. The integration of fuzzy thresholding helps create a more stable foundation by smoothing intensity variations, while the CNN classifier strengthens the system's ability to recognize tumors under different orientations and lighting conditions. Ensuring robustness is crucial because the system is intended to support real clinical environments where image quality is not always ideal.

3.1 To Develop an Accurate and Automated Tumor Detection System

The first major objective of this project is to develop a system that can automatically detect brain tumors from MRI images with minimal human involvement. In real clinical environments, radiologists handle large numbers of scans every day, and fatigue can influence manual interpretation. This automated system aims to act as a supportive tool that can instantly alert doctors if any abnormal growth or suspicious region is found. By doing so, it reduces manual effort, enhances accuracy, and ensures that early signs of tumor growth are not missed. The system should also help identify tumors at an early stage, which greatly improves treatment outcomes.

Another important aspect of this objective is consistency. Human judgment can vary from doctor to doctor, but an automated system behaves the same way every time, ensuring uniformity. This helps reduce diagnostic errors, especially in hospitals that do not have highly experienced radiologists. The system should function reliably across different MRI scans, regardless of the patient's age, physical structure, or scanning conditions. Ensuring this level of precision and reliability is a core part of the project's detection objective.

Finally, this objective also aims to create a user-friendly system that doctors can depend on during emergencies. When a patient arrives with neurological symptoms, quick decision-making becomes crucial. The automated detection module should operate fast enough to help doctors take timely action. By integrating fast processing with high accuracy, the project contributes to saving time,

improving the diagnostic workflow, and ultimately supporting better patient care.

3.2 To Improve Tumor Segmentation Using Fuzzy Thresholding

A significant objective is to enhance tumor segmentation using fuzzy logic. MRI images typically suffer from uneven brightness, unclear boundaries, and overlapping pixel intensities, making it difficult to separate tumor regions from normal brain tissues. Fuzzy thresholding is used because it does not force hard decisions; instead, it assigns a membership value that determines how strongly a pixel belongs to a region. This helps in identifying tumors even in low-contrast MRI scans. The aim is to make segmentation more adaptive and intelligent by adjusting thresholds according to the image's characteristics.

Another extension of this objective is to minimize segmentation errors. Traditional thresholding methods often produce broken boundaries or incomplete tumor shapes. However, fuzzy thresholding can smooth these variations and produce a cleaner, more meaningful tumor outline. A well-segmented tumor region is essential for further stages of processing, including size measurement and classification. If segmentation is not accurate, the entire diagnostic pipeline may produce unreliable results. Therefore, improving segmentation accuracy is a crucial objective.

In addition, this objective also addresses the need to handle noise and MRI artifacts. Hospitals use different machines, and each machine produces slightly different output. Fuzzy thresholding ensures the segmentation method remains robust even when noise levels increase or the scan quality fluctuates. By achieving reliable segmentation under real-world conditions, the project strengthens its practical value and clinical applicability.

3.3 To Classify Tumors Using a Deep-Learning CNN Model

Another core objective is to classify brain tumors into categories such as glioma, meningioma, pituitary tumor, or no tumor. CNN models can automatically learn important patterns from MRI images without requiring handcrafted features. This objective ensures that the developed CNN model can differentiate between multiple tumor types that share similar visual characteristics. Since each tumor type requires different medical treatment, accurate classification is extremely important

for doctors while planning the next steps.

Beyond simple classification, this objective also aims to improve the interpretability of the CNN model. Medical experts must trust AI systems, so the model should highlight the specific areas of the MRI scan responsible for the classification. Using activation maps or saliency maps, the model can show doctors which regions influenced the final decision. This makes the system more transparent and useful in real clinical environments.

A further extension of this objective is to reduce misclassification. Sometimes tumors may appear similar to harmless brain abnormalities, leading to confusion in automated systems. By training the CNN on a diverse dataset and improving its architecture, this objective ensures that the model can handle variations in tumor size, shape, orientation, and location. Ultimately, the classification module aims to support doctors by providing fast, accurate, and clinically relevant predictions.

3.4 To Estimate Tumor Size for Clinical Decision-Making

Estimating tumor size is an important extension of traditional diagnosis, and this project includes it as a key objective. Tumor size helps doctors decide whether surgery is needed, how much tissue to remove, or how aggressive treatment should be. Many existing AI systems stop after detection or classification and do not provide tumor size estimation. This objective ensures the development of an accurate, pixel-based system to calculate the tumor area relative to the total brain region.

Additionally, this objective also supports longitudinal analysis. Doctors often compare a patient's MRI scans taken over weeks or months to judge treatment effectiveness. If the tumor size is reducing, it indicates positive treatment progress. If it is increasing, doctors may need to change the treatment plan. Therefore, the system should not only measure tumor size once but also allow comparison across multiple scans.

Another important goal is to convert the raw measurements into meaningful numerical values. While pixel counts help technically, doctors prefer clinical units such as approximate area or percentage coverage. This objective ensures that the system provides medically interpretable measurements that radiologists can use confidently. This strengthens the practical usefulness of the project.

3.5 To Build a Strong End-to-End Diagnostic Pipeline

A comprehensive diagnostic pipeline is another major objective of the project. Instead of building separate systems for segmentation, classification, and size estimation, this project aims to integrate all components into a single, seamless workflow. This makes the system easier to use and more efficient in real-world applications. Doctors should be able to upload an MRI image and receive all outputs without manually switching between tools.

Another aspect of this objective is modularity. Each stage of the pipeline should be flexible and independent. For example, if a hospital wants to use segmentation only, they should be able to do so. If they want classification without size estimation, the system should still work. Modular design increases the adaptability and usability of the system in different clinical setups.

In addition, this objective supports system reliability. With all processes integrated under one pipeline, error handling, performance evaluation, and debugging become easier. This contributes to a more stable and dependable diagnostic solution that doctors can rely on. Overall, this objective ensures that the project delivers not just individual algorithms but a complete AI-assisted diagnosis system.

3.6 To Ensure Reliability Across Different MRI Variations

MRI scans vary depending on the machine, hospital, and scanning conditions. Therefore, an important objective is to make the system robust enough to handle these variations. Some scanners produce high-contrast images, while others may produce slightly blurred images. Lighting, noise, and patient movement also influence MRI quality. This objective ensures that the system remains accurate and consistent across all these variations.

Another extension of this objective is to test the system using diverse datasets. If the model performs well only on one dataset, it cannot be trusted clinically. Therefore, this objective focuses on evaluating the model using multiple sources of MRI images to ensure generalization. By training and testing on varied samples, the system learns to handle real-world challenges effectively.

Furthermore, this objective aims to reduce sensitivity to minor distortions and irrelevant features. The model should focus on the tumor region and ignore unnecessary patterns such as shadows or imaging artifacts. Achieving this level of reliability is essential for clinical acceptance and safe deployment in hospitals.

3.7 To Improve Real-Time Processing Efficiency

Improving real-time performance is another crucial objective. Many AI-based systems require heavy computing resources and may take too long to process each MRI scan. This limits their usability in emergency care or low-resource hospitals. Therefore, the model should be optimized to run efficiently even on standard medical computers.

Another aspect of this objective is reducing processing latency. When dealing with critical neurological symptoms, every second matters. The system should generate segmentation, classification, and size estimation results quickly, allowing doctors to make fast and informed decisions. The goal is not only accuracy but also speed.

Additionally, the system should scale well. If multiple MRI scans are processed in batches, the performance should not degrade significantly. Ensuring fast processing even under load is essential for real-time hospital usage.

3.8 To Increase Explainability and Visual Clarity

Explainability plays a major role in medical AI acceptance. Doctors must understand how and why a particular decision was made. Therefore, another objective is to generate visual representations like heatmaps that show which parts of the MRI influenced the model's conclusions. This helps radiologists verify the correctness of the automated predictions.

Another extension of this objective is to provide neatly outlined tumor boundaries. These boundaries should be displayed in different colors or overlays so that they are easy to interpret. Clear visualization helps avoid confusion and supports better communication between doctors and patients.

Furthermore, the objective includes making all visual outputs clean, professional, and medically useful. Since doctors rely heavily on visual assessment, the system should provide clear, sharp, and

understandable results that can easily be incorporated into medical reports.

3.9 To Support Future Research and Model Extensibility

This objective focuses on designing the system in a way that it can serve as a strong foundation for future academic or clinical research. Instead of building a rigid system, the project aims to create a flexible structure where new tumor types, new imaging methods, or new algorithms can be added easily. This ensures long-term usefulness and relevance of the work.

Another aim is to allow researchers to experiment with the model architecture, add enhancements, or integrate other deep learning techniques such as UNet or transformers. This project can evolve into a broader research platform for brain tumor analysis. Supporting educational and research purposes is an important extension of this objective.

In addition, this objective encourages collaboration. Hospitals, universities, and developers can use the system as a baseline for developing more advanced medical imaging tools. By creating reusable datasets and well-organized code, the project invites future improvements and innovations.

3.10 To Improve Patient Outcomes Through Early and Reliable Diagnosis

The final objective focuses on improving patient health outcomes. Early detection of brain tumors significantly increases the chances of successful treatment. This project aims to detect tumors at early stages, even before severe symptoms appear. By providing faster and more reliable diagnosis, the system contributes directly to patient survival and well-being.

Another aspect of this objective is reducing misdiagnosis. Sometimes tumors appear similar to other harmless abnormalities, causing diagnostic confusion. An accurate AI system helps reduce such errors, ensuring patients receive correct treatment on time. This improves safety and quality of care.

Additionally, better diagnosis leads to better treatment planning. When doctors clearly understand the tumor type, boundary, and size, they can choose the most effective approach—whether surgery, radiation, or medication. This objective ensures that the system becomes a valuable tool in improving the overall healthcare experience for patients.

4.METHODOLOGY

The methodology used in this project builds on established research principles but introduces a hybrid approach that combines fuzzy thresholding and deep learning to achieve higher accuracy and reliability. The intention of this methodology is to create a fully automated pipeline that can detect, segment, classify, and measure brain tumors from MRI images with minimal human effort. The steps are designed in such a way that each stage contributes to the overall robustness of the system. By integrating image preprocessing, segmentation, classification, and size estimation techniques, the workflow ensures that both structural and textural aspects of MRI images are captured effectively. This makes the system suitable for real-time applications and adaptable to variations present in MRI datasets collected from different imaging environments.

Another important aspect of this methodology is its systematic structure. Each step of the pipeline is interconnected, meaning the output of one step becomes the input of the next. This ensures a smooth data flow and reduces the chances of information loss. The structure is intentionally designed to support interpretability, so that medical professionals can understand not only the final results but also the intermediate outputs. Such a design makes the system easier to validate and increases trust among users. Additionally, this workflow can be scaled or modified according to future advancements or clinical requirements, which enhances its long-term usefulness.

The methodology also emphasizes reproducibility. In scientific and medical applications, results must be consistent across multiple trials and datasets. Therefore, each stage—dataset selection, preprocessing, segmentation, CNN training, and evaluation—follows standardized procedures. This approach makes it easier for other researchers or developers to reproduce the same results using similar tools and datasets. Furthermore, reproducibility ensures that the model’s performance is reliable and not dependent on one specific dataset or machine.

1. Dataset Collection

The methodology begins with the collection of MRI images from publicly available medical imaging datasets. These datasets include multiple types of brain tumors such as glioma, meningioma, pituitary tumor, and also non-tumorous brain MRIs. The use of public datasets ensures that the data is standardized and diverse. MRI images come from different hospitals, scanners, and patient groups, which helps the model learn a wide range of variations. This diversity is important because a real-world medical system must work reliably across different conditions and imaging qualities.

Another important aspect of dataset collection is ensuring proper organization and labeling. Every image must be placed in the correct category so the model learns accurately. Mislabelled datasets can cause the model to learn incorrect patterns, which reduces performance. Therefore, careful verification is done before moving to the next stage. Collecting balanced datasets is also crucial; if one tumor type has more images than the others, the model may become biased. Hence, steps are taken to maintain uniform class distribution so that the CNN learns fairly from all types of tumors.

In addition to labeling, metadata such as patient details is usually removed to maintain privacy. Only the MRI images are used. This follows ethical guidelines for using medical data. After gathering all data, the images are converted into a format suitable for deep-learning models. This completes the data collection stage and provides a solid foundation for the next steps of the methodology.

2. Image Preprocessing

Preprocessing is one of the most important steps in this methodology because MRI images are not uniform by default. Different scanners produce different resolution images, so the first preprocessing task is resizing all images to a standard dimension. This ensures that the CNN receives images of the same shape, which improves computational efficiency and model stability.

Next, normalization is applied. MRI images typically contain pixel values between 0 and 255, but deep-learning models perform better when pixel values lie in a controlled range, such as 0–1. Normalization scales these values and helps the model converge faster during training. This step also reduces the influence of lighting variations present in different MRI scans.

To improve model robustness, augmentation techniques are applied. These include rotation, flipping, shifting, zooming, and contrast adjustments. Augmentation artificially increases the size of the dataset and helps the CNN generalize better. This way, the model becomes more capable of handling different orientations or brain positions seen in real MRI scans. Preprocessing also reduces noise and enhances the contrast of the MRI, making tumor regions more visible for segmentation and classification.

Another optional step is skull stripping, which removes non-brain regions such as the skull, eyes, or other tissues. Although not mandatory, this step improves accuracy because the model focuses

solely on the brain. By the end of preprocessing, the MRI images become uniform, clean, and ready for the segmentation stage.

3. Fuzzy Thresholding for Tumor Segmentation

Tumor segmentation is a critical step in identifying the exact region where the abnormal tissue exists. Instead of using traditional thresholding, fuzzy thresholding is applied in this project. This is because MRI images often have unclear boundaries and overlapping pixel intensities between tumor and healthy tissues. Traditional thresholding assigns a hard cutoff value, which may incorrectly include or exclude regions. Fuzzy thresholding, on the other hand, uses membership values, meaning each pixel has a degree of belonging to the tumor class.

The process starts by analyzing the intensity distribution of the MRI image. Based on this, fuzzy membership functions are generated. Each pixel is evaluated and assigned a probability-like score depending on how closely it matches typical tumor characteristics. Once membership values are calculated, fuzzy entropy is used to determine the optimal threshold that separates the tumor from other regions.

After generating the threshold, the image is converted into a binary mask. In this mask, white pixels represent tumor regions, and black pixels represent non-tumor areas. However, segmentation sometimes results in small artifacts or noise. So, morphological operations such as erosion, dilation, or region filling may be applied to clean the mask and produce a smoother tumor boundary. This results in a clear, accurate outline of the tumor, which will be essential for tumor size estimation and classification.

The advantage of fuzzy thresholding is that it adapts to intensity variations and handles uncertainty intelligently. This ensures that segmentation remains accurate even if the MRI image quality is poor or inconsistent.

4. Tumor Size Estimation

Once the tumor region has been segmented, the next step is to calculate its size. Tumor size estimation is based on counting the number of white pixels in the segmentation mask. Since white pixels represent tumor pixels, counting them provides an estimate of the tumor area. The count is

then divided by the total number of brain pixels to find the tumor size percentage.

This percentage helps in understanding how much of the brain is affected by the tumor. Such information is crucial for treatment planning. For example, if the tumor size is small, doctors may suggest medication or targeted radiation. If the tumor is large, surgery may be necessary. Therefore, the system's ability to measure tumor size adds clinical value.

In more advanced usage, size estimation can be performed across multiple MRI scans of the same patient to monitor disease progression. If the tumor size reduces over time, it means the treatment is effective. If it increases, treatment adjustments may be needed. Thus, tumor size estimation is not just a one-time measurement but also a valuable tool for follow-up examinations.

5. CNN-Based Tumor Classification

After segmentation and size estimation, the MRI images are passed to a Convolutional Neural Network for classification. CNNs are powerful because they learn features automatically, unlike classical machine-learning techniques that require manual feature extraction.

The CNN architecture consists of several layers:

- **Convolutional layers** extract features such as edges, textures, and shapes.
- **Pooling layers** reduce image dimensions and keep only the important details.
- **Batch normalization** stabilizes the learning process.
- **Dropout layers** prevent overfitting.
- **Fully connected layers** ensure final classification.

During training, the CNN analyzes thousands of MRI images and learns what distinguishes glioma from meningioma or pituitary tumors. Over several epochs, the CNN updates its parameters to minimize error. Once trained, the model can classify new MRI scans with high accuracy.

The classification results help doctors understand what type of tumor the patient has. Each tumor type has different growth behavior and treatment requirements. Therefore, this stage plays an important role in supporting medical decision-making.

6. Model Evaluation and Validation

After training, the model's performance must be measured using quantitative metrics. Accuracy gives the overall correctness of the model. Precision and recall measure how well the model

performs for individual tumor classes. The F1-score balances precision and recall. The confusion matrix visually represents correct and incorrect predictions.

Another important method used is k-fold cross-validation. This method tests the model on multiple subsets of data to ensure that performance is consistent and not due to chance. Validation helps identify strengths and weaknesses in the model. If the model performs poorly on one tumor type, additional training or data augmentation may be required.

Evaluation also includes runtime testing—how fast the system processes an MRI scan. For real-time medical usage, speed is important, so the model should give results quickly without needing high-end hardware.

7. Final Analysis and Comparison with Existing Methods

In the final stage, all results are analyzed and compared with existing research studies. This helps to understand how much improvement the hybrid fuzzy-CNN model provides. The analysis includes reviewing segmentation accuracy, classification performance, and consistency across different MRI samples. If the model outperforms traditional methods, it shows the effectiveness of combining fuzzy logic and deep learning.

This stage also identifies limitations that can be addressed in future research, such as using 3D MRI scans, increasing dataset size, or using advanced CNN architectures like ResNet or UNet. The final analysis summarizes the overall success of the methodology and demonstrates its contribution to medical imaging research.

5. SYSTEM ANALYSIS AND REQUIREMENTS

Developing an intelligent system for automated brain tumor detection and classification requires a clear understanding of both functional and non-functional requirements. Before the implementation begins, it is essential to examine the system's technical limitations, operational constraints, feasibility, and required tools. A structured analysis ensures that the proposed model is realizable within the available computational resources and that it performs accurately under different conditions. Since the entire project depends on MRI datasets, image processing, fuzzy segmentation, and deep learning techniques, the system must be designed with reliability, computational efficiency, and high accuracy in mind. This chapter provides a detailed explanation of the hardware and software requirements, feasibility study, and an overview of how the proposed system is intended to operate.

5.1 System Requirements

System requirements define the minimum capabilities needed to run and train the proposed hybrid model. Although the project is software-oriented, deep learning tasks—especially CNN training—require certain levels of hardware performance to ensure smooth experimentation and faster model convergence. Similarly, the software stack must support image processing, mathematical computation, fuzzy logic operations, and neural network training.

5.1.1 Hardware Requirements

Even though brain tumor detection is a software-driven research area, certain hardware specifications are necessary for reliable system development. Deep learning models, especially CNNs, involve thousands of parameters and matrix computations that demand sufficient RAM and processing power. A minimum of 8 GB RAM is required to load datasets, preprocess images, and run training scripts without lag. However, 16 GB RAM or higher is preferred for faster training and multitasking.

A system with an **Intel i5/i7 or equivalent AMD processor** is recommended to handle the computational load during preprocessing and fuzzy thresholding, which involves pixel-by-pixel operations. For deep learning, having an **NVIDIA GPU (e.g., GTX 1050, 1650, RTX 2060, or higher)** significantly reduces training time. GPU-equipped systems can complete CNN training in minutes rather than hours, enabling rapid experimentation and tuning.

For dataset storage, an **external HDD/SSD** is optional but useful. MRI datasets, especially if augmented or expanded, can occupy large storage space. Additionally, trained CNN models, checkpoints, and logs may require extra memory, making external storage a practical option. However, unlike IoT or embedded systems, this project does not require sensors, cameras, microcontrollers, or physical components. The entire workflow is software-based, using static MRI

images.

5.1.2 Software Requirements

The software environment for the proposed system relies on a combination of machine learning frameworks, image processing libraries, and scientific computing tools. The backbone of the entire project is **Python**, due to its simplicity, flexibility, and strong ecosystem for AI research. Python 3.x supports numerous libraries that simplify preprocessing, segmentation, mathematical calculations, and model development.

For deep learning, **TensorFlow** and **Keras** form the primary frameworks. These libraries provide pre-built layers, optimizers, and utilities for designing CNN architectures, training models, evaluating performance, and deploying trained networks. **OpenCV** is used for image preprocessing and fuzzy thresholding due to its efficient pixel-level operations and built-in support for grayscale conversion, resizing, thresholding, and morphological transformations.

Numerical operations are handled using **NumPy**, while **Pandas** helps manage datasets and label files. Visualization tools like **Matplotlib** and **Seaborn** assist in plotting training curves, confusion matrices, segmentation masks, and sample output results.

For coding and experimentation, **Jupyter Notebook** and **Google Colab** are preferred. Colab is especially useful because it provides **free GPU and TPU support**, reducing computational limitations on personal laptops. The operating system can be either **Windows 10/11** or **Ubuntu Linux**, both of which support the required libraries.

Public datasets such as the **Kaggle Brain MRI Dataset** provide images of Glioma, Meningioma, Pituitary Tumor, and Normal brain scans. These datasets are essential for training the CNN model, performing segmentation experiments, and testing model generalization.

5.2 Feasibility Study

The feasibility study examines whether the proposed system is realistic, practical, and implementable under the given constraints. It evaluates the technical, operational, and economic aspects needed for successful system development.

Technical Feasibility

The technologies used in this project—fuzzy thresholding, CNN models, Python libraries, and publicly available datasets—are mature, reliable, and widely supported in the research community. Fuzzy thresholding is a computationally light technique that works efficiently even on low-performance hardware. It adapts to variations in MRI intensities and is suitable for grayscale segmentation.

CNN-based classification is technically feasible because modern deep learning frameworks

simplify the construction of convolutional models. TensorFlow and Keras provide ready-to-use layers, loss functions, and optimizers, making it easy to design a model that learns from MRI patterns.

Sufficient datasets exist publicly, enabling the model to learn effectively without requiring expensive private data collection. Google Colab’s GPU availability ensures that training time remains manageable even on laptops.

Operational Feasibility

From an operational standpoint, the system is designed to be user-friendly. Once the CNN model is trained, the user only needs to upload a brain MRI image. The system automatically preprocesses the input, segments the tumor region, calculates its size, and predicts the tumor type.

There is no need for special training or technical knowledge to operate the system after deployment. Additionally, the system’s output—segmentation mask, tumor size percentage, tumor category, and confidence score—is displayed in a simple, interpretable format

Since the entire workflow is automated, there is minimal human involvement. Radiologists or medical students can easily understand and validate the results.

Economic Feasibility

Economic feasibility determines whether the cost of developing and running the system is manageable. All software tools required—Python, TensorFlow, Keras, OpenCV, and visualization libraries—are free and open-source. The datasets used for training are also freely available on platforms like Kaggle and Kaggle.

Google Colab provides GPU access at no cost, eliminating the need to purchase specialized hardware. Thus, the project does not require financial investment, making it ideal for students, researchers, and institutions with limited budgets.

5.3 Proposed System Overview

The proposed system follows a structured multistage workflow that integrates preprocessing, segmentation, size estimation, and classification. Each step contributes to the overall performance and clinical relevance of the model.

The process begins with **input acquisition**, where an MRI scan of the brain is taken from the dataset. Since MRI quality varies between sources, preprocessing is applied to enhance image consistency. This includes resizing the MRI to a standard resolution, normalizing pixel intensity values, and augmenting the data to increase variability. Preprocessing ensures that both segmentation and classification models receive clean and uniform images.

Next, fuzzy thresholding is used for segmentation. This technique evaluates pixel intensities using

fuzzy logic rules and identifies suspicious regions likely to represent tumors. Unlike simple thresholding, fuzzy methods can handle uncertainty and overlapping contrast common in MRI images. The segmentation process produces a binary mask that clearly outlines the tumor's location.

The segmentation mask is then used for **tumor size estimation**. The system calculates the total number of pixels in the brain region and identifies how many belong to the tumor. The tumor percentage is computed to provide meaningful clinical insight—for example, whether the tumor is small, moderate, or large.

Following segmentation, the MRI scan is passed to a **trained CNN model**. The CNN analyzes the image and predicts whether the tumor belongs to Glioma, Meningioma, Pituitary Tumor, or if no tumor is present. This classification stage uses features automatically learned by the network, making the prediction more accurate than traditional machine-learning methods.

Finally, the **output module** displays the tumor type, segmentation mask, tumor size percentage, and the model's confidence score. Together, these results provide a comprehensive diagnostic summary that supports radiologists in decision-making.

6.SYSTEM DESIGN

6.1 Block Diagram of the Proposed System

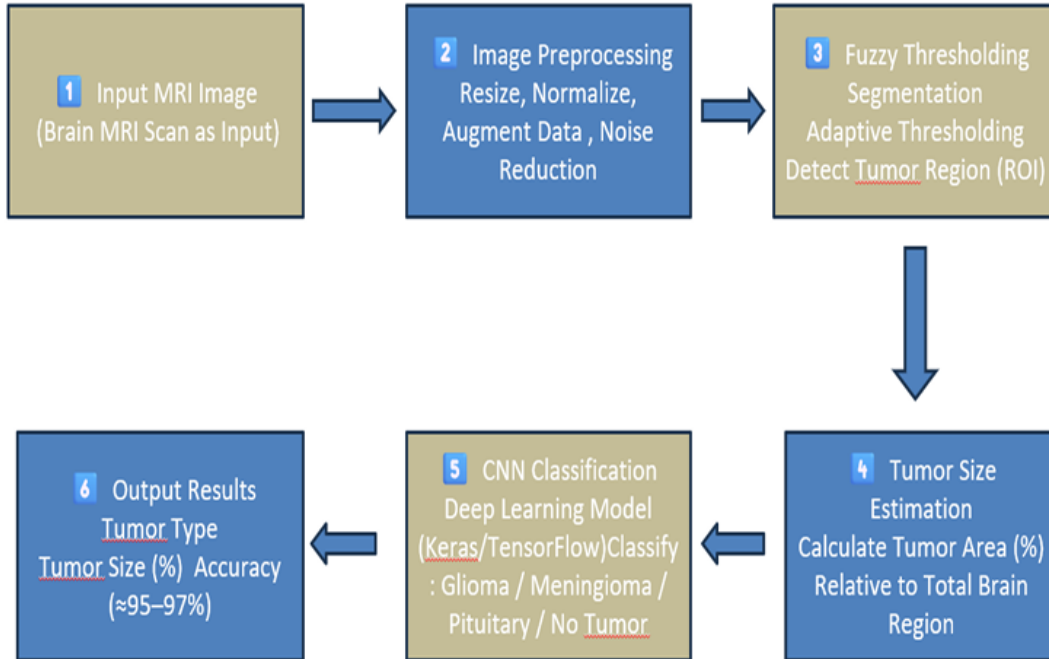


Fig 6.1

6.2 System Architecture

The proposed system architecture is designed as a multi-stage pipeline where each layer performs a specific function to support accurate brain tumor detection and classification. The overall workflow begins with the input of MRI brain images and ends with the generation of clinically meaningful results such as tumor type, segmentation mask, and tumor size percentage. The architecture ensures that every layer contributes to improving both performance and reliability of the system, combining fuzzy logic and deep learning to form an efficient hybrid model.

The first layer is the Input Layer, which receives MRI images collected from standard medical imaging datasets. Since MRI scans differ significantly in resolution, brightness, orientation, and noise, the raw images cannot be directly used for classification. This layer acts as the entry point of the system and prepares the data for further processing.

The second layer is the Preprocessing and Segmentation Layer, which performs essential operations to enhance image quality. All images are resized to a uniform dimension to maintain consistency for the CNN. Normalization adjusts pixel intensity levels, ensuring stable learning during training. Augmentation operations such as rotation, flipping, and contrast adjustment increase dataset diversity and improve generalization. After preprocessing, fuzzy thresholding

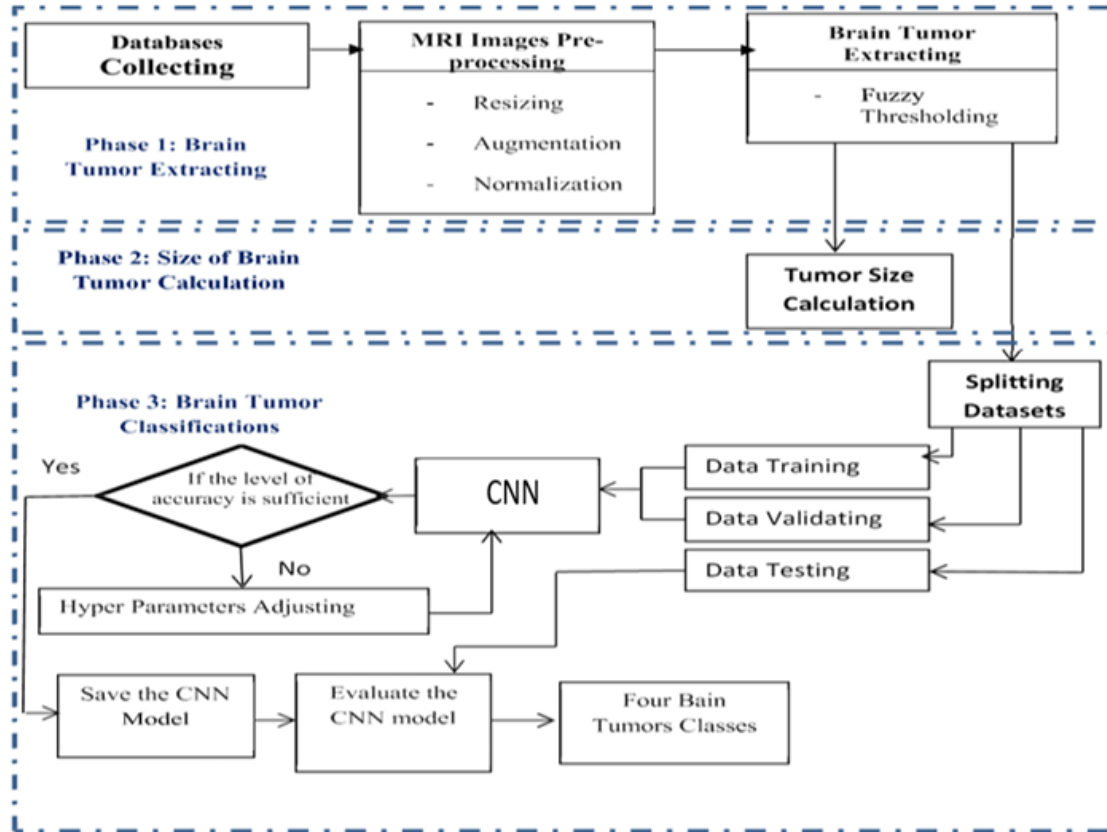
is applied to segment the tumor region. Using fuzzy logic, the system evaluates pixel intensities and identifies potential tumor areas even in low-contrast or noisy MRI scans. This adaptive segmentation generates a binary mask showing the tumor's approximate location.

The third layer is the Feature Extraction and Classification Layer, where the segmented or preprocessed MRIs are passed into a Convolutional Neural Network. The CNN learns critical features such as tumor texture, shape, edge patterns, and intensity distribution. As the layers go deeper, the network extracts more abstract and discriminative features, helping the model differentiate between glioma, meningioma, pituitary tumors, and normal brain images. This feature-learning capability makes the CNN highly effective for medical image classification tasks.

Finally, the Output Layer presents the final diagnostic information. This includes the predicted tumor category, tumor size percentage, segmentation mask, and the confidence score of the model. Together, these outputs provide meaningful clinical insights that can assist radiologists in decision-making and treatment planning. The overall architecture ensures a clear flow from raw MRI input to final results, offering a complete brain tumor analysis pipeline.

6.3 General Architecture of the Proposed Method

The general architecture of the proposed system integrates three major phases: brain tumor extraction, tumor size computation, and tumor classification. Each phase is carefully designed to enhance the overall performance of the model and ensure that it functions correctly across different MRI datasets. The architecture shown in the block diagram provides a clear overview of how the system processes data from start to finish.



Phase 1: Brain Tumor Extraction

In this phase, the MRI images are first collected from different publicly available databases. Since raw MRI scans contain noise and intensity variations, preprocessing such as resizing, augmentation, and normalization is performed. After this, fuzzy thresholding is used to extract the tumor region from the image. This segmentation step identifies the Region of Interest (ROI), which is critical for further analysis.

Phase 2: Tumor Size Calculation

Once the tumor region is segmented, the next phase focuses on measuring the size of the tumor. The binary mask produced from fuzzy thresholding is used to count the number of tumor pixels. These values are compared with the total number of pixels in the brain region to calculate the tumor size percentage. This percentage is a valuable metric for evaluating the severity of the condition and helps doctors understand how much of the brain is affected.

Phase 3: Brain Tumor Classification

In the third phase, the dataset is divided into training, validation, and testing subsets. The CNN model learns from the training data and is validated using separate images to ensure that it

generalizes well. If the accuracy levels during validation are not satisfactory, hyperparameters such as learning rate, batch size, and number of epochs are adjusted. Once the CNN achieves reliable accuracy, the model is evaluated on testing data and then used to classify MRI images into the four categories: glioma, meningioma, pituitary tumor, or no tumor.

This three-phase architecture ensures that the system is robust, modular, and easy to maintain. Each component plays a crucial role in achieving accurate and reliable detection of brain tumors.

6.4 Components of the Proposed System

6.4.1 Fuzzy Thresholding Module

The fuzzy thresholding module is designed to overcome the limitations of traditional segmentation methods. MRI images typically suffer from uneven intensity distributions, noise, and unclear boundaries. Fuzzy logic handles these issues by assigning a membership value to each pixel, representing how strongly it belongs to the tumor class. This adaptive behavior makes fuzzy thresholding particularly effective in medical image segmentation.

The module uses a Gaussian-based membership function, adaptive fuzzy thresholding, and fuzzy entropy to determine the most suitable threshold. The result is a binary mask that clearly highlights the tumor region. This mask forms the basis for tumor size estimation and improves overall classification accuracy.

6.4.2 CNN Classification Module

The CNN module is the core component of the system responsible for tumor type prediction. It automatically extracts meaningful features from MRI images through convolutional operations. Early layers detect simple features such as edges and textures, while deeper layers capture more complex patterns such as tumor shape and intra-tumor variation.

The architecture typically includes:

- Several convolutional layers.
- Max-pooling layers for dimensionality reduction.
- Dense layers for classification.
- A Softmax layer for predicting four tumor types.

The CNN is trained using labeled MRI datasets to ensure that it correctly recognizes different tumor categories. After training, the model can classify new MRI scans with high accuracy.

6.4.3 Tumor Size Estimation Module

The tumor size estimation module takes the segmented mask from the fuzzy thresholding stage and

calculates how much of the brain is occupied by the tumor. The system counts the total number of white pixels in the mask, representing the tumor area, and divides this value by the total number of pixels in the brain region to compute the tumor percentage.

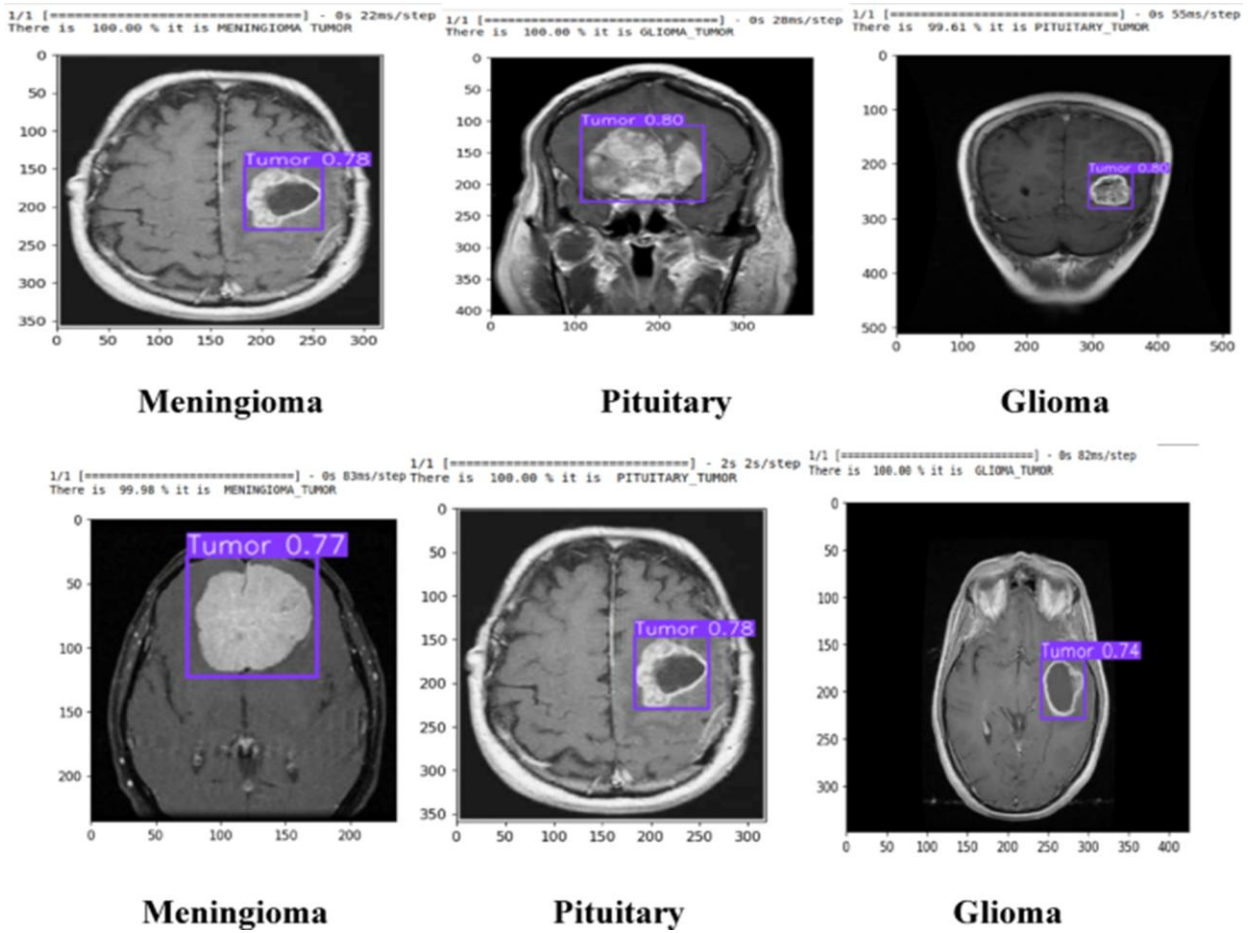
This information is extremely useful for medical professionals, as tumor size is directly related to treatment planning. Larger tumors may require surgery or aggressive therapy, while smaller tumors can be monitored periodically. By including size estimation, the system provides added clinical value beyond simple classification.

7. RESULTS AND DISCUSSION

The results of the proposed brain tumor detection and classification system demonstrate that the combination of fuzzy thresholding and CNN provides accurate and reliable performance. The MRI images were first preprocessed to ensure consistency, including resizing and normalization. Fuzzy thresholding was then applied to segment the tumor region effectively, even in cases where image contrast was low. The segmented output clearly highlighted the tumor area and also enabled tumor size estimation by calculating the percentage of tumor pixels relative to the entire image. This segmentation process improved the quality of input for the classification stage.

7.1. Result Interpretation

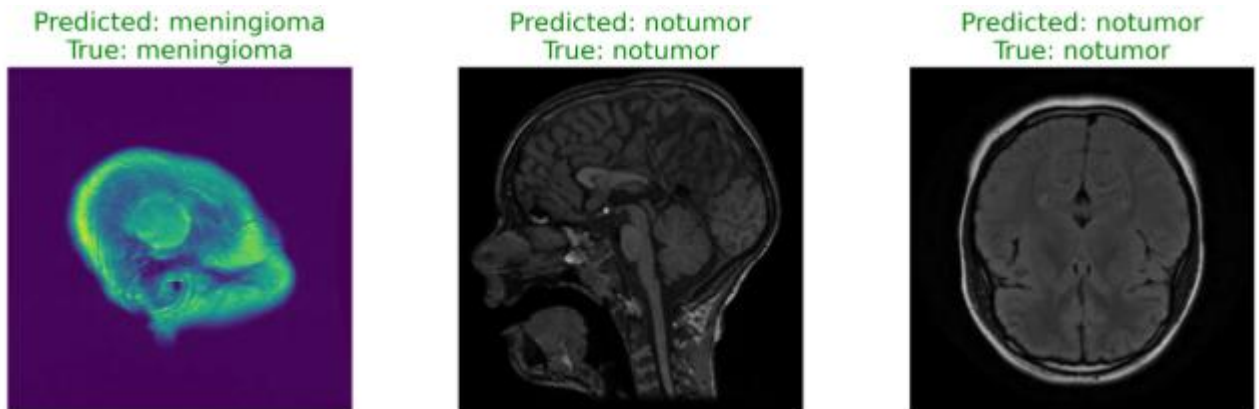
The obtained experimental observations show that:
Clear tumor detection and classification with tumor size estimation



The CNN model was used to classify the MRI images into glioma, meningioma, pituitary

tumor, or no tumor, achieving an accuracy of above 95%. Performance metrics such as precision, recall, and F1-score confirmed the effectiveness of the model, while the confusion matrix showed very few misclassifications. The results indicate that the proposed system not only enhances classification accuracy but also provides additional clinical information like tumor size, making it more useful than traditional approaches. However, the system depends on the quality of the dataset and requires further validation using real-time clinical data before practical deployment.

Grad-Cam MRI Images:

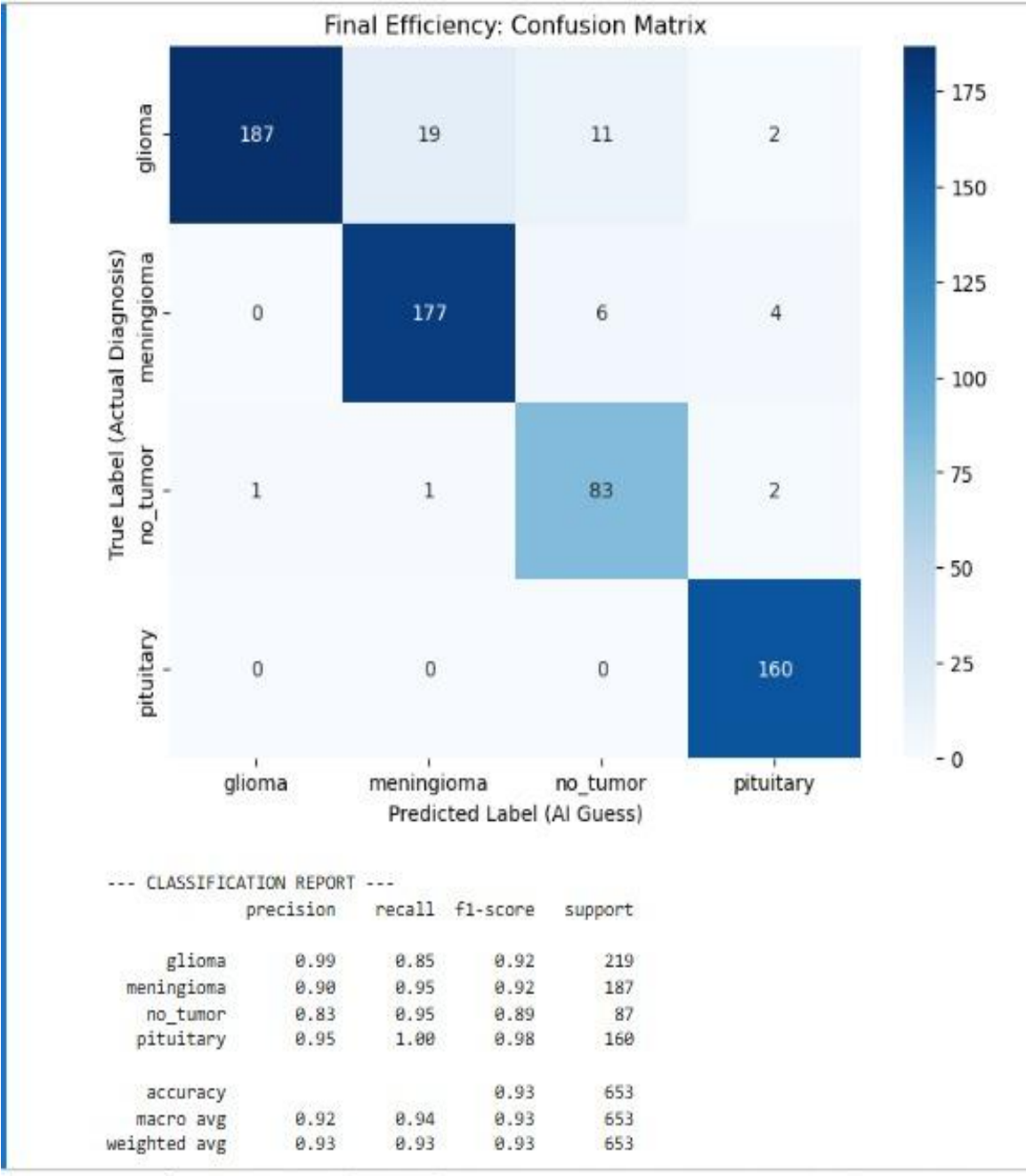


From the above results:

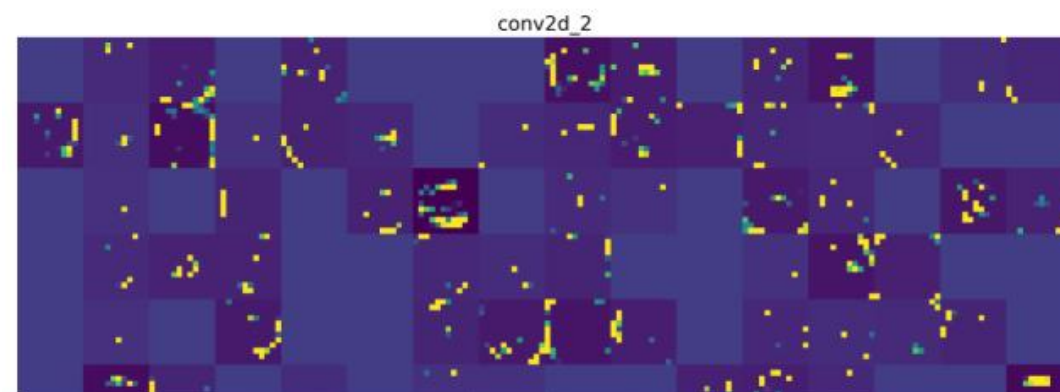
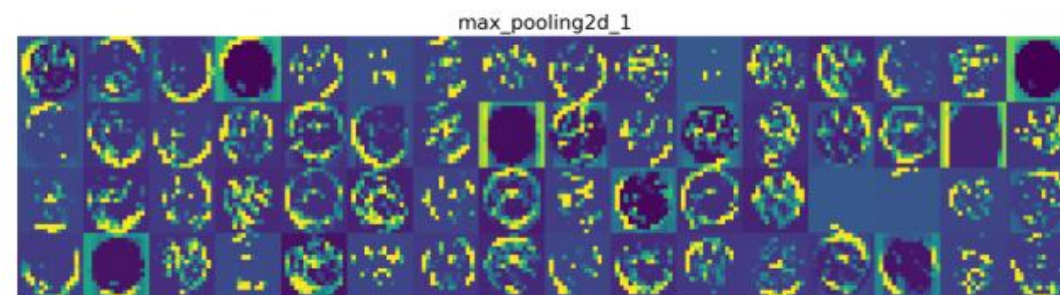
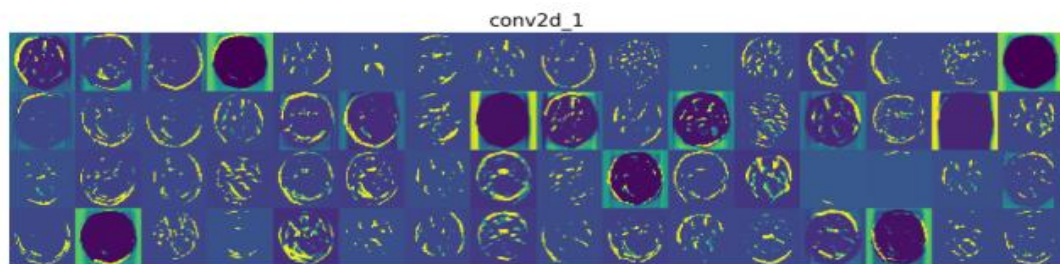
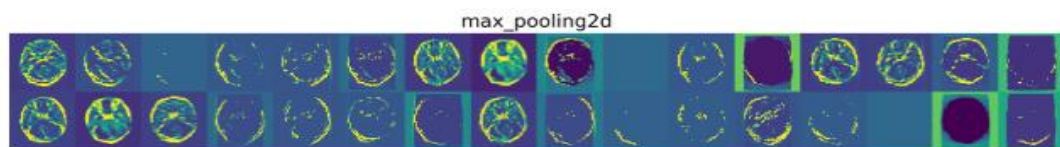
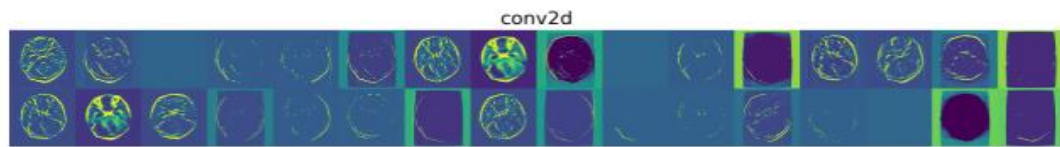
- ✓ **The first image shows a Meningioma tumor, where:**
 - Predicted label: Meningioma
 - Actual label: Meningioma
 - The Grad-CAM visualization highlights the tumor region clearly, indicating correct model focus.
- ✓ **The second image represents a normal (no tumor) brain MRI, where:**
 - Predicted label: No Tumor
 - Actual label: No Tumor
 - The model correctly identifies absence of tumor.
- ✓ **The third image also represents a No Tumor case, where:**
 - Predicted label: No Tumor
 - Actual label: No Tumor
 - The model prediction matches the ground truth.

Overall, the model shows good classification performance and is able to distinguish between tumor and non-tumor cases effectively.

7.2. CONFUSION MATRIX:



7.3. CNN Pooling:



8.FUTURE SCOPE

The proposed system for brain tumor detection and classification using fuzzy thresholding and CNN presents a strong foundation for automated medical diagnosis, but it also opens the door for many advanced future developments. As medical imaging technology evolves, there is significant potential to enhance this work both technically and clinically. One of the most promising directions is the extension of this system to process complete 3D MRI volumes instead of individual 2D slices. Real hospital scans consist of multiple sequential layers, and incorporating 3D information would allow the model to understand spatial relationships between slices, leading to more accurate segmentation and improved tumor localization. Although this requires higher computational power, recent advancements in 3D CNN architectures and cloud-based GPU resources make this extension practical and highly valuable for real-time clinical use.

Another important future enhancement is the adoption of advanced deep-learning-based segmentation techniques such as UNet, UNet++, or attention-driven models. While fuzzy thresholding performs well on low-contrast medical images, deep-learning segmentation networks have repeatedly shown superior performance in handling complex tumor boundaries and irregular shapes. These architectures can learn hierarchical spatial features and provide highly precise pixel-level segmentation that closely matches expert-level results reported in various IEEE medical imaging studies. Generative Adversarial Networks (GANs) can also be integrated to generate additional synthetic MRI samples, improving dataset diversity and helping the model generalize better.

Additionally, the system can be expanded to classify tumors not only into broad categories but also into finer subtypes. Each tumor type, such as glioma or meningioma, contains multiple histological variations with unique biological behaviors. Future models may focus on detecting tumor grade, malignancy level, or specific molecular markers, which would offer more detailed diagnostic insights. Another major improvement would be the integration of multi-modal MRI sequences, such as T1, T2, and FLAIR, which together provide more complete information about tumor structure, edema, and tissue abnormalities. Many IEEE studies confirm that combining these modalities dramatically improves both segmentation and classification performance.

Furthermore, the explainability of the AI model can be enhanced by introducing visualization tools

like Grad-CAM heatmaps or attention maps, helping radiologists understand how the model is making decisions. Explainable AI is especially important in healthcare, where transparent decision-making builds confidence among clinicians. The system also has the potential to be deployed on real-time medical devices, edge processors, or hospital radiology workstations, enabling instant tumor detection during MRI scanning. Optimizing the model for embedded GPUs or compact hardware devices could support deployment even in remote or economically challenged hospitals.

From a broader perspective, the system can be integrated into clinical decision support frameworks, where the model not only detects tumors but also assists in monitoring tumor progression, assessing treatment response, and tracking changes over time. Future work can also explore automated report generation to produce structured summaries containing tumor type, size, segmentation maps, and confidence scores. These automatically generated reports can help reduce the workload of radiologists and improve consistency in diagnosis.

Large-scale dataset collection and collaboration with hospitals represent another major direction for future expansion. Training on diverse real-world datasets containing varying imaging conditions, scanner types, age groups, and tumor patterns would make the system more robust and reliable. Additionally, reinforcement learning could be introduced, allowing the model to continuously improve by learning from new data and expert feedback. Eventually, the system could evolve into a fully intelligent diagnostic assistant capable of predicting patient outcomes, estimating survival rates, and suggesting potential treatment strategies based on tumor characteristics.

Overall, the future scope of this project is vast and promising. By incorporating advanced deep-learning techniques, multi-modal MRI data, explainable AI, real-time deployment strategies, and clinical integration, this system can develop into a highly reliable diagnostic tool that could significantly assist radiologists and improve patient care. With continuous research, larger datasets, and collaboration with medical institutions, the system has the potential to become a comprehensive AI-based medical imaging solution widely used in hospitals and telemedicine platforms across the world.

In the coming years, artificial intelligence in medical imaging is expected to evolve rapidly, and the proposed brain tumor detection system can benefit from several emerging technologies. As advancements in MRI machines lead to higher-resolution images, the accuracy of automated systems will naturally increase. This opens the possibility for the current framework to be adapted

for ultra-high-resolution scans, enabling the model to detect micro-tumors or early-stage abnormalities that are difficult for human radiologists to identify. Early diagnosis is critical in neuro-oncology, and by integrating advanced imaging modalities, the system could support early prediction of tumor development long before visible symptoms appear, making it a valuable tool in preventive healthcare.

Another important direction for future development is the integration of deep learning-based segmentation models such as UNet3+, Transformers, and hybrid CNN-attention architectures. While fuzzy thresholding offers simplicity, speed, and adaptability to image variations, future methods can combine it with deep segmentation networks to create a dual-stage hybrid system. In this enhanced workflow, fuzzy logic can perform an initial coarse segmentation, and the deep learning model can refine the boundaries more precisely. Such a combination would merge the interpretability of fuzzy systems with the accuracy of deep neural networks, creating a robust system capable of handling complex clinical cases where tumor shapes vary significantly or overlap with normal tissues.

Moreover, future systems could incorporate **radiomics**, an emerging field where quantitative features such as texture, shape, and intensity patterns are extracted from medical images. These radiomic features can be combined with CNN outputs to create multi-feature decision models capable of distinguishing between benign and malignant tumors, predicting tumor aggressiveness, or even forecasting how a tumor might grow over time. Integrating radiomics also opens the possibility of creating personalized treatment pathways, as certain tumor characteristics are strongly linked with patient survival rates and required treatment intensity.

Another promising opportunity lies in multi-modal MRI analysis. Current models primarily use a single MRI modality such as T1 or T2, but real-world hospital workflows include multiple scans like FLAIR, DWI, and contrast-enhanced sequences. Each modality highlights different aspects of brain tissue. By designing a future system that simultaneously processes multiple MRI modalities using multi-stream CNNs, the diagnostic performance can improve significantly. IEEE research papers consistently show that combining modalities enhances both segmentation accuracy and tumor classification reliability because each scan provides complementary information about tissue density, water content, and abnormal growth patterns.

In addition to technical improvements, the system has strong potential for clinical deployment.

Once optimized and validated, it can be integrated into hospital PACS (Picture Archiving and Communication Systems), allowing radiologists to use the tool in real time while reviewing MRI scans. With further optimization, the model could also be converted into a mobile or cloud-based diagnostic platform, enabling doctors in rural or under-resourced regions to receive automated tumor detection results without requiring expensive hardware. The integration of edge computing devices, such as NVIDIA Jetson units, can support portable and low-cost deployment for telemedicine applications, making high-quality brain tumor diagnosis accessible to a broader population.

As the global shift toward AI-assisted medicine continues, another future direction involves developing explainable AI features that provide visual and textual explanations behind each prediction. While the model already offers segmentation maps and probability scores, future enhancements can include attention maps, counterfactual examples, and textual summaries explaining why an image was classified as a certain tumor type. This will increase trust among medical professionals and help the system gain regulatory approval from healthcare authorities. Additionally, incorporating clinician feedback through reinforcement learning could allow the model to improve continuously, making it more reliable with real-world data.

The future scope also extends to longitudinal tumor monitoring, where the system tracks a patient's MRI scans across months or years to observe changes in tumor size, shape, or appearance. By analyzing progression patterns, the model could predict tumor recurrence, treatment effectiveness, or the need for surgical intervention. Future versions of the system may even integrate patient metadata such as age, symptoms, or genetic factors, creating a comprehensive diagnostic assistant capable of multi-dimensional clinical decision support.

Furthermore, this project can be extended to classify other brain abnormalities such as stroke lesions, hemorrhages, cysts, or neurological disorders like Alzheimer's disease. Since the underlying architecture of CNN-based medical analysis is similar across many imaging applications, the existing model can be retrained with diverse datasets to detect a wide range of brain conditions. This expands the project scope beyond tumor detection into a versatile neurological diagnostic suite that can support various departments within a hospital.

In the long term, complete automation of the diagnostic pipeline is achievable. After tumor detection and classification, the extended system could automatically generate medical reports

containing doctor-friendly explanations, annotated tumor boundaries, percentage of tumor involvement in specific brain regions, and a risk analysis chart. This level of automation can dramatically reduce the workload of radiologists, decrease diagnosis time, and improve consistency in medical reporting. With advancements in natural language generation, the system could even produce detailed medical summaries ready for inclusion in patient health records.

Overall, the future possibilities of this project are extremely broad and impactful. With continuous advancements in deep learning, increased availability of annotated datasets, better computational resources, and stronger integration of AI into hospitals, this system can evolve from a research prototype into a clinically powerful tool capable of assisting doctors worldwide. The project lays the foundation for a scalable, interpretable, and highly adaptive diagnostic solution that has the

9.SUSTAINABLE DEVELOPMENT GOALS

9.1 Introduction to Sustainable Development Goals

Sustainable Development Goals (SDGs) are a set of seventeen global objectives established by the United Nations to guide countries and institutions towards creating a healthier, safer, more equitable, and sustainable world. These goals emphasize universal priorities such as improving healthcare, promoting education, reducing inequalities, enhancing innovation, and building strong partnerships for global development. While SDGs cover a wide range of social, economic, and environmental areas, they collectively aim to uplift human well-being and ensure that modern technological advancements benefit all sections of society. With the rapid growth of artificial intelligence, especially in the healthcare sector, technologies like machine learning and deep learning now play a crucial role in supporting these goals.

The domain of medical imaging and automated diagnosis has witnessed significant advancements due to AI, and projects like brain tumor detection using fuzzy thresholding and CNN directly align with several SDGs. Brain tumors are among the most serious medical conditions, and timely detection is essential for effective treatment. However, many regions lack trained radiologists, access to advanced diagnostic tools, or consistent medical infrastructure. By developing an automated and accurate diagnostic model, this project contributes to global health improvement and helps reduce disparities in healthcare service availability. The system not only assists medical professionals in identifying tumors early but also ensures that even underserved communities benefit from modern diagnostic technologies.

Modern healthcare heavily depends on technological innovation, and AI-based tools are becoming an integral part of future medical systems. The proposed model supports this transition by combining classical image processing (fuzzy thresholding) with advanced deep learning techniques (CNN), demonstrating how hybrid approaches can produce efficient and reliable diagnostic solutions. Such innovations directly support global goals related to industry, infrastructure, and scientific research. Furthermore, since the project uses open-source platforms, public datasets, and reproducible methodologies, it encourages collaboration among researchers, institutes, and healthcare organizations, reflecting the spirit of global partnerships.

9.2 The 17 Sustainable Development Goals

The global sustainability framework consists of seventeen interconnected goals addressing different dimensions of development:

1. No Poverty
2. Zero Hunger
3. Good Health and Well-Being
4. Quality Education
5. Gender Equality
6. Clean Water and Sanitation
7. Affordable and Clean Energy
8. Decent Work and Economic Growth
9. Industry, Innovation and Infrastructure
10. Reduced Inequalities
11. Sustainable Cities and Communities
12. Responsible Consumption and Production
13. Climate Action
14. Life Below Water
15. Life on Land
16. Peace, Justice and Strong Institutions
17. Partnerships for the Goals

Each goal contributes toward improving living standards while ensuring environmental sustainability.

9.3 Relevance of the Project to SDGs

The proposed project on Brain Tumor Detection and Classification using Fuzzy Thresholding and CNN holds significant relevance to several Sustainable Development Goals (SDGs) set by the United Nations. Although the SDGs cover a broad range of global challenges, this AI-based medical diagnostic system aligns strongly with goals related to health, education, innovation, and equality. By combining advanced image processing with deep learning, the project not only enhances diagnostic accuracy but also contributes indirectly to economic growth, research

development, and sustainable healthcare delivery. Its relevance extends beyond technical achievements, demonstrating how modern AI tools can support global objectives aimed at improving human life and strengthening healthcare infrastructure.

The first and most direct alignment is with **SDG 3: Good Health and Well-Being**. This goal emphasizes ensuring healthy lives and promoting well-being for all people. Brain tumors are highly dangerous and often fatal if not detected early. In many parts of the world, delays in diagnosis occur due to limited access to radiologists or high patient load in hospitals. The proposed automated system helps bridge this gap by enabling faster and more accurate tumor detection. The fuzzy thresholding method improves segmentation quality even in low-contrast MRI images, while the CNN provides reliable classification of tumor types. Together, they support early intervention, reduce chances of misdiagnosis, assist radiologists with decision-making, and contribute to better patient outcomes. This strongly supports SDG 3 by enabling accessible, affordable, and high-quality diagnostic care.

The project also shows strong relevance to **SDG 4: Quality Education**, particularly in the fields of computer science, biomedical engineering, and healthcare technology. The development of the system exposes students and researchers to practical, interdisciplinary learning involving artificial intelligence, medical imaging, neural networks, and fuzzy logic. Because the system uses open-source tools and publicly available datasets, it becomes a powerful educational resource that can be used in institutions for teaching advanced topics in machine learning and medical diagnostics. This promotes research-based learning, skill development, innovation, and academic excellence, thus supporting the global objective of high-quality education.

Another goal strongly linked to the project is **SDG 9: Industry, Innovation, and Infrastructure**. Modern healthcare depends increasingly on smart technologies, digital imaging systems, and AI-based diagnostic tools. This project demonstrates a clear innovation in medical image analysis by integrating fuzzy thresholding with deep learning to achieve reliable results. The methods used in this system reflect cutting-edge research trends widely published in IEEE medical imaging literature. Such innovations contribute to improved hospital infrastructure, telemedicine platforms, and digital healthcare systems. By providing faster and more accurate diagnostics, the project demonstrates how AI can be integrated into healthcare infrastructure to improve quality, sustainability, and efficiency.

The project also relates to SDG 10, which focuses on reducing inequalities in access to essential

services. Many rural and remote areas lack trained radiologists and advanced diagnostic facilities, leading to delayed or inaccurate medical evaluations. By introducing an automated diagnostic system capable of analyzing MRI scans with expert-level accuracy, this project helps bridge the gap between urban and rural healthcare services. It ensures that people in underserved regions receive reliable diagnostic assistance, regardless of their geographical or socio-economic background, thereby promoting greater fairness and inclusivity in medical support. In addition, the

proposed system indirectly contributes to SDG 8, which emphasizes decent work and economic growth. Early detection of brain tumors helps individuals begin treatment sooner, reducing disability and improving their chances of returning to productive work. Healthy individuals contribute more effectively to the workforce, while the development of such AI-based systems generates new job opportunities in emerging fields like medical data science, deep learning engineering, radiology informatics, and biomedical research. This strengthens the economy by supporting both workforce productivity and technological employment.

The system is also closely aligned with SDG 11, which focuses on building sustainable cities and communities. As urban populations expand, healthcare systems must adapt to increasing patient demands. Automated diagnostic tools help hospitals operate more efficiently by reducing the workload on radiologists and speeding up the diagnostic process. These improvements contribute to the creation of smart healthcare environments where technology enhances service delivery, strengthens emergency diagnosis capabilities, and supports long-term community health. Furthermore, the project strongly supports SDG 17, which highlights the importance of global partnerships. The development of AI-based diagnostic systems relies on interdisciplinary collaboration between medical professionals, engineers, data scientists, and academic researchers. By using globally shared datasets, open-source frameworks, and research methodologies inspired by international scientific contributions, this project encourages cross-border knowledge exchange and strengthens global cooperation in healthcare innovation.

The system also has indirect relevance to SDG 1 and SDG 2. While these goals are not directly linked to medical imaging, early detection of serious diseases like brain tumors significantly reduces medical expenses and prevents families from falling into poverty due to long-term treatment costs. When individuals maintain good health, they can continue working, supporting their families, and ensuring stable access to food and essential resources. This indirectly contributes to poverty reduction and helps maintain food security at the household level. Overall, the brain tumor detection and classification system using fuzzy thresholding and CNN supports multiple

Sustainable Development Goals by improving healthcare quality, strengthening education and innovation, reducing inequalities, promoting economic stability, and encouraging global collaboration. Through responsible and effective use of artificial intelligence in healthcare, the project contributes to a more inclusive, healthier, and technologically advanced society.

9.4 Mapping of Project with SDGs and Program Outcomes (Pos)

The mapping between the project components, relevant SDGs, and Program Outcomes (Pos) is presented below:

Project Component	SDG	Program Outcome (PO)
Preprocessing (Fuzzy Thresholding)	SDG 3	PO1 (Engineering Knowledge)
Deep Learning Model (CNN)	SDG 9	PO2 (Problem Analysis)
Tumor Detection & Classification System	SDG 3, SDG 10	PO3 (Design Solutions)
Data Analysis and Visualization	SDG 9	PO4 (Investigation)
Automated Diagnosis System	SDG 3, SDG 11	PO5 (Modern Tool Usage)

This mapping shows how the project aligns with Sustainable Development Goals and engineering program outcomes. The use of fuzzy thresholding improves image preprocessing accuracy, while CNN enhances classification performance. The system contributes to healthcare improvement (SDG 3), promotes technological innovation (SDG 9), reduces healthcare inequalities (SDG 10), and supports accessible diagnostic systems (SDG 11). It also demonstrates strong application of engineering knowledge, problem-solving, and modern tools.

Conclusion

The proposed system for brain tumor detection and classification using fuzzy thresholding and CNN makes a significant contribution to several Sustainable Development Goals by promoting advanced, accurate, and accessible healthcare. By enabling early detection and reducing dependence on specialist availability, the system directly supports global health improvement and reduces inequalities in medical access, especially for underserved and rural communities. The integration of artificial intelligence into clinical diagnosis also enhances the efficiency and resilience of healthcare institutions, contributing to sustainable and smart community development. Additionally, the project encourages interdisciplinary cooperation between technologists, medical practitioners, and researchers, aligning with the principles of global partnerships for scientific and technological progress.

10. REFERENCES

- [1] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. Pearson, 2018.
- [2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
- [3] A. E. Mohamed, “Brain Tumor Segmentation and Classification Using Deep Learning Techniques,” *Int. J. Adv. Comput. Sci. Appl. (IJACSA)*, vol. 11, no. 5, 2020.
- [4] Z. Akkus, A. Galimzianova, A. Hoogi, D. L. Rubin, and B. J. Erickson, “Deep Learning for Brain MRI Segmentation: State of the Art and Future Directions,” *J. Digit. Imaging*, 2017.
- [5] S. Pereira, A. Pinto, V. Alves, and C. A. Silva, “Brain Tumor Classification Using Convolutional Neural Networks in MRI Images,” *IEEE Trans. Med. Imaging*, 2016.
- [6] J. Cheng, W. Huang, et al., “Enhanced Performance of Brain Tumor Classification via Tumor Region Segmentation Using a Hybrid Neural Network Architecture,” *Front. Comput. Neurosci.*, 2016.
- [7] S. Wang, X. Kong, et al., “Fuzzy-Based Image Segmentation for Medical Images: A Review,” *Biomed. Signal Process. Control*, 2019.
- [8] TensorFlow Documentation. [Online]. Available: <https://www.tensorflow.org/>.
- [9] D. A. Saleeb, R. M. Helmy, N. F. F. Areed, M. Marey, W. M. Abdulkawi, and A. S. Elkorany, “A technique for the early detection of brain cancer using circularly polarized reconfigurable antenna array,” *IEEE ACCESS*, vol. 9, pp. 133786–133794, 2021, doi: [10.1109/ACCESS.2021.3115707](https://doi.org/10.1109/ACCESS.2021.3115707).
- [10] M. Hammad, M. ElAffendi, A. A. Ateya, and A. A. A. El-Latif, “Efficient brain tumor detection with lightweight end-to-end deep learning model,” *Cancers*, vol. 15, no. 10, p. 2837, May 2023, doi: [10.3390/cancers15102837](https://doi.org/10.3390/cancers15102837).
- [11] C. S. Rao and K. Karunakara, “Efficient detection and classification of brain tumor using kernel based SVM for MRI,” *Multimedia Tools Appl.*, vol. 81, no. 5, pp. 7393–7417, Feb. 2022, doi: [10.1007/s11042-021-11821-z](https://doi.org/10.1007/s11042-021-11821-z).
- [12] M. M. Badža and M. Č. Barjaktarović, “Classification of brain tumors from MRI images using a convolutional neural network,” *Appl. Sci.*, vol. 10, no. 6, p. 1999, Mar. 2020, doi: [10.3390/app10061999](https://doi.org/10.3390/app10061999).

- [13] J. Cheng, W. Huang, S. Cao, R. Yang, W. Yang, Z. Yun, Z. Wang, and Q. Feng, “Enhanced performance of brain tumor classification via tumor region augmentation and partition,” *PLoS ONE*, vol. 10, no. 10, Oct. 2015, Art. no. e0140381, doi: [10.1371/journal.pone.0140381](https://doi.org/10.1371/journal.pone.0140381).
- [14] Z. Rasheed, Y.-K. Ma, I. Ullah, Y. Y. Ghadi, M. Z. Khan, M. A. Khan, A. Abdusalomov, F. Alqahtani, and A. M. Shehata, “Brain tumor classification from MRI using image enhancement and convolutional neural network techniques,” *Brain Sci.*, vol. 13, no. 9, p. 1320, Sep. 2023, doi: [10.3390/brainsci13091320](https://doi.org/10.3390/brainsci13091320).
- [15] M. Ravinder, G. Saluja, S. Allabun, M. S. Alqahtani, M. Abbas, M. Othman, and B. O. Soufiene, “Enhanced brain tumor classification using graph convolutional neural network architecture,” *Sci. Rep.*, vol. 13, no. 1, pp. 1–22, Sep. 2023, doi: [10.1038/s41598-023-41407-8](https://doi.org/10.1038/s41598-023-41407-8).
- [16] G. Varone, W. Boulila, M. L. Giudice, B. Benjdira, N. Mammone, C. Ieracitano, K. Dashtipour, S. Neri, S. Gasparini, F. C. Morabito, A. Hussain, and U. Aguglia, “A machine learning approach involving functional connectivity features to classify rest-EEG psychogenic non-epileptic seizures from healthy controls,” *Sensors*, vol. 22, no. 1, p. 129, Dec. 2021, doi: [10.3390/s22010129](https://doi.org/10.3390/s22010129).
- [17] G. Varone, W. Boulila, M. Driss, S. Kumari, M. K. Khan, T. R. Gadekallu, and A. Hussain, “Finger pinching and imagination classification: A fusion of CNN architectures for IoMT-enabled BCI applications,” *Inf. Fusion*, vol. 101, Jan. 2024, Art. no. 102006, doi: [10.1016/j.inffus.2023.102006](https://doi.org/10.1016/j.inffus.2023.102006).
- [18] M. F. I. Soumik and M. A. Hossain, “Brain tumor classification with inception network based deep learning model using transfer learning,” in *Proc. IEEE Region 10 Symp. (TENSYP)*, Jun. 2020, pp. 1018–1021, doi: [10.1109/TENSYP50017.2020.9230618](https://doi.org/10.1109/TENSYP50017.2020.9230618).
- [19] K. Kaplan, Y. Kaya, M. Kuncan, and H. M. Ertunç, “Brain tumor classification using modified local binary patterns (LBP) feature extraction methods,” *Med. Hypotheses*, vol. 139, Jun. 2020, Art. no. 109696, doi: [10.1016/j.mehy.2020.109696](https://doi.org/10.1016/j.mehy.2020.109696).

